

Exercise 5 of module 1004D on "Theoretical spectroscopy for periodic systems"**(Time-dependent) perturbation theory**

Compute the dynamical polarizability of the H atom depending on the frequency ω . The perturbation operator is hereby given as

$$\hat{W}(t) = E_0(t)\hat{\mu}_z\theta(t), \quad \theta(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (1)$$

The time evolution of the wave function is described by the time evolution operator $\hat{U}(t)$,

$$\Psi(t) = \hat{U}(t)\Psi(t=0). \quad (2)$$

- Insert the ansatz for $\Psi(t)$ into the time-dependent Schrödinger equation and determine the time evolution for $\hat{U}(t)$. Solve the equation for a time-independent perturbation.
- Select the following ansatz for the wave function,

$$|\Psi(t)\rangle = \exp(-i\hat{H}^{(0)}t)|\Psi_I(t)\rangle, \quad (3)$$

with $|\Psi_I\rangle$ representing the wave function in the interaction picture. An operator $\hat{A}$ in the interaction picture is given as

$$\hat{A}_I = e^{-i\hat{H}^{(0)}t}\hat{A}e^{i\hat{H}^{(0)}t}. \quad (4)$$

Show that the time evolution operator fulfills the following equation,

$$i\frac{\partial}{\partial t}\hat{U}_I(t) = \hat{W}_I(t)\hat{U}_I(t). \quad (5)$$

- Transfer the differential equation for $\hat{U}_I(t)$ into an integral equation and solve this equation up to first order in the external potential. Start with $\hat{U}_I^{(0)}(t=0) = \mathbf{1}$.
- Compute the expectation value for the dipole moment μ_z according to

$$\langle\hat{\mu}_z(t)\rangle = \langle\Psi_I|\hat{U}_I^\dagger(t)\hat{\mu}_z\hat{U}_I(t)|\Psi_I\rangle. \quad (6)$$

- The polarizability α_{zz} is defined as

$$\alpha_{zz}(t, t') = \left. \frac{\delta\langle\hat{\mu}_z(t)\rangle}{\delta E(t')} \right|_{E(t')=0}. \quad (7)$$

Show that $\alpha_{zz}(t, t')$ only depends on $(t - t')$.

- Compute the Fourier transform of the polarizability. Introduce a convergence factor of the form $\exp(-\varepsilon t)$ with $\varepsilon > 0$.
- Draw the real and imaginary part of the frequency dependent polarizability as a function of the frequency. Discuss the influence of ε .

Time evolution of expectation values

The expectation values of a time-dependent N -electron wave function $\Psi(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N, t)$ and a time-independent operator $\hat{A}$ is time dependent.

- a** Assuming that the time evolution of Ψ is described by the time-dependent Schrödinger equation, prove that

$$\frac{d\langle\Psi|\hat{A}|\Psi\rangle}{dt} = i\langle\Psi|[\hat{H}, \hat{A}]|\Psi\rangle. \quad (8)$$

$[\hat{H}, \hat{A}] = \hat{H}\hat{A} - \hat{A}\hat{H}$ defines the commutator of operators $\hat{H}$ and $\hat{A}$.

- b** Derive Eq. (8) for the position and momentum operator $\langle\mathbf{r}\rangle$ and $\langle\mathbf{p}\rangle$. How do these equations differ from the classical equations of motion?
- c** The electron density can be written as the expectation value over the density operator $\langle\delta(\mathbf{r} - \mathbf{r}_1)\rangle$. Compute the commutator between the molecular Hamiltonian $\hat{H}$ and the density operator, $[\hat{H}, \delta(\mathbf{r} - \mathbf{r}_1)]$.
- d** Use results of a)-c) to derive an equation for the time evolution of the electron density.

Gradient for Hartree-Fock theory

The Hartree-Fock energy is given as

$$E_{\text{HF}} = \sum_{\mu\nu} D_{\mu\nu} H_{\mu\nu} + \frac{1}{2} \sum_{\mu\nu} D_{\mu\nu} \sum_{\kappa\lambda} D_{\kappa\lambda} (\mu\nu|\kappa\lambda) + V_{\text{NN}}, \quad (9)$$

with

$$(\mu\nu|\kappa\lambda) = (\mu\nu|\kappa\lambda) - \frac{1}{2}(\mu\lambda|\kappa\nu), \quad (10)$$

$$D_{\mu\nu} = 2 \sum_i C_{\mu i} C_{\nu i}. \quad (11)$$

- a** Derive the equations for the nuclear gradient by taking the straight-forward total derivative according to

$$\frac{dE_{\text{HF}}}{d\zeta} = \frac{\partial E_{\text{HF}}}{\partial \zeta} + \sum_{\mu i} \frac{\partial E_{\text{HF}}}{\partial C_{\mu i}} \frac{dC_{\mu i}}{d\zeta}. \quad (12)$$

- b** Set up a Lagrange functional taking orbital orthogonality as additional constraint to restore stationarity with respect to the MO coefficients. Derive the nuclear gradient, now using Lagrange's method of undetermined multipliers. Compare the equations of **b)** with **a)**.